A QCQS ALGEBRAIC SPACE HAS A SURJECTIVE FINITE SCHEME COVER

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3798-3815, following the statement at 3785-3796.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Integral scheme over a decent space. Decent Spaces, Lemma decent-spaces-lemma-there-is-a-scheme-integral-over (cross-chapter, cited at ~3799-3801). Surjective integral (hence finite) Y -> X with local factors through U.
- Integral limit is a limit of finite fp morphisms. lemma-integral-limit-finite-and-finite-presentation (same chapter, cited at ~3803-3805). Y = lim Y_i with Y_i -> X finite and of finite presentation.
- Limits that are schemes and descended opens. lemma-limit-is-scheme (same chapter, cited at ~3806-3807); lemma-descend-opens (same chapter, cited at ~3808-3810). Y_i is a scheme for large i; affine opens V_{i,j} in Y_i recovering V_j.
- Morphisms appear at finite stages. proposition-characterize-locally-finite-presentation (same chapter, card L26289, cited at ~3811-3813). V_j -> U over X comes from V_{i,j} -> U for i large.

Source reference. Original reference 09YB: focused-seeds/geometry/S9-spaces-limits-there-is-a-scheme-finite-over/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0827: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1295–1295. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 09YA: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2939–2939. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SR: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1435–1435. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 04AK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 694–694. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S.*

(1) *There exists a surjective finite morphism $Y \to X$ of finite presentation where Y is a scheme,*
(2) *given a surjective étale morphism $U \to X$ we may choose $Y \to X$ such that for every $y \in Y$ there is an open neighbourhood $V \subset Y$ such that $V \to X$ factors through U.*

Proof. Part (1) is the special case of (2) with $U = X$. Let $Y \to X$ be as in Decent Spaces, Lemma 09YB. Choose a finite affine open covering $Y = \bigcup V_j$ such that $V_j \to X$ factors through U. We can write $Y = \lim Y_i$ with $Y_i \to X$ finite and of finite presentation, see Lemma 09YA. For large enough i the algebraic space Y_i is a scheme, see Lemma 07SR. For large enough i we can find affine opens $V_{i,j} \subset Y_i$ whose inverse image in Y recovers V_j, see Lemma 0827. For even larger i the morphisms $V_j \to U$ over X come from morphisms $V_{i,j} \to U$ over X, see Proposition 04AK. This finishes the proof. □